Seminar 1:

Classroom problems and worked solutions

25117 - Econometrics

Universitat Pompeu Fabra

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The issue: what can data tell us about training?

- A university offers job training. We want to understand students' employment and earnings, and decide what the evidence says about the programme.
- We will describe uncertainty, study relationships, and compare ways to estimate an average.
- For every problem: identify the relevant population, write the calculation, and interpret the result in words and units.

Classroom problems

Each problem states its own population and assumptions. Work through the questions before drawing a policy conclusion. Worked solutions follow each problem.

Problem 1: Who receives weekly job earnings?

Select one of four students with equal probability. Weekly job earnings, measured in hundreds of euros, are

Student	s_1	s_2	s_3	s_4
$W(s)$	0	0	1	3

1. Construct the probability mass function of W. Check that its probabilities sum to one.
2. Write and sketch the complete c.d.f. $F_W(w) = \Pr(W \leq w)$.
3. Find $F_W(0)$, $F_W(1)$, $\Pr(0 < W \leq 1)$, and $\Pr(W \geq 1)$. Why do s_1 and s_2 correspond to one value of W?

Solution 1: The earnings probability distribution

1.

w	0	1	3
$\Pr(W = w)$	1/2	1/4	1/4

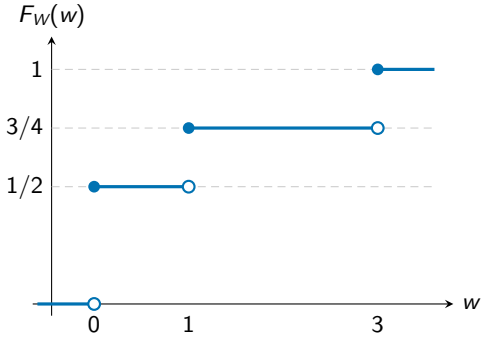
$$\frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1.$$

Solution 1: The earnings c.d.f. (continued)

2.

$$F_W(w) = \begin{cases} 0, & w < 0, \\ \frac{1}{2}, & 0 \le w < 1, \\ \frac{3}{4}, & 1 \le w < 3, \\ 1, & w \ge 3. \end{cases}$$

At each jump, the filled dot gives the value of the right-continuous c.d.f.



3.

$$F_W(0) = \frac{1}{2}, \qquad F_W(1) = \frac{3}{4},$$

$$\Pr(0 < W \le 1) = F_W(1) - F_W(0) = \frac{1}{4}, \qquad \Pr(W \ge 1) = 1 - \Pr(W = 0) = \frac{1}{2}.$$

Distinct students s_1 and s_2 both have $W = 0$.

Problem 2: How long until a training interview?

Waiting time T, measured in hours, is continuous with density

$$f_T(t) = \begin{cases} ct, & 0 \le t \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

1. Find c so that f_T is a valid density. Sketch the density.
2. Derive the complete c.d.f. $F_T(t)$, including values outside $[0, 1]$.
3. Calculate $\Pr(T \le 1/2)$ and $\Pr(1/4 < T \le 3/4)$. Shade the corresponding areas under the density.
4. Find $\Pr(T = 3/4)$. Does a density greater than one at $t = 3/4$ violate a probability rule?

Solution 2: A density and its cumulative probability

1.

$$1 = \int_{-\infty}^{\infty} f_T(t)\,dt = \int_0^1 ct\,dt = \left[\frac{ct^2}{2}\right]_0^1 = \frac{c}{2} \quad \Longrightarrow \quad \boxed{c = 2}.$$

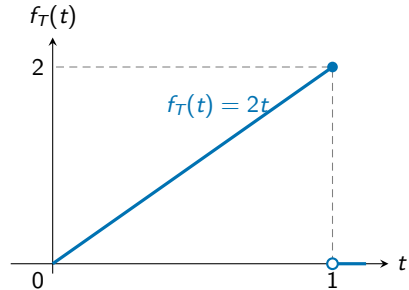
2. For $0 \le t \le 1$,

$$F_T(t) = \int_0^t 2u\,du = t^2.$$

Thus the complete c.d.f. is

$$F_T(t) = \begin{cases} 0, & t < 0, \\ t^2, & 0 \le t \le 1, \\ 1, & t > 1. \end{cases}$$

1. Density sketch

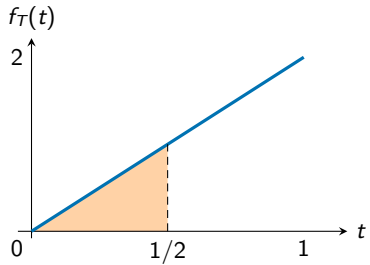


The density is nonnegative and its total area is one; its integral gives accumulated probability.

Solution 2: Probabilities are shaded areas (continued)

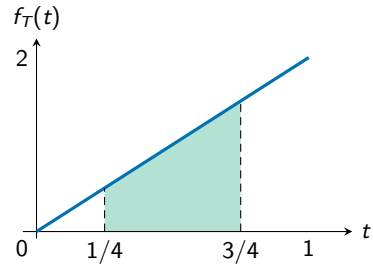
3.

Waiting at most half an hour



$$\Pr(T \le \frac{1}{2}) = \int_0^{1/2} 2t\,dt = \left[t^2\right]_0^{1/2} = \boxed{\frac{1}{4}}.$$

Waiting between 15 and 45 minutes



$$\Pr(\frac{1}{4} < T \le \frac{3}{4}) = \frac{9}{16} - \frac{1}{16} = \boxed{\frac{1}{2}}.$$

4. $\Pr(T = 3/4) = 0$: a single point has zero area. The height $f_T(3/4) = 3/2$ per hour is allowed; probabilities are integrals, not density heights.

Problem 3: What are typical weekly earnings?

Use the earnings distribution from Problem 1:

w (hundreds of euros)	0	1	3
$\Pr(W = w)$	1/2	1/4	1/4

1. Compute $\mu_W = \mathbb{E}(W)$ and $\mathbb{E}(W^2)$.
2. Calculate $\mathrm{Var}(W)$ from the weighted squared deviations. Check it using $\mathbb{E}(W^2) - [\mathbb{E}(W)]^2$.

Solution 3: Average earnings and variance

1.

$$\mathbb{E}(W) = \frac{1}{2}(0) + \frac{1}{4}(1) + \frac{1}{4}(3) = 1, \qquad \mathbb{E}(W^2) = \frac{1}{4}(1) + \frac{1}{4}(9) = \frac{5}{2}.$$

2.

Weighted squared deviations

$$\begin{aligned}\mathrm{Var}(W) &= \frac{1}{2}(0-1)^2 + \frac{1}{4}(1-1)^2 \\ &\quad + \frac{1}{4}(3-1)^2 \\ &= \frac{1}{2} + 0 + 1 = \boxed{\frac{3}{2}}.\end{aligned}$$

Second-moment shortcut

$$\begin{aligned}\mathrm{Var}(W) &= \mathbb{E}(W^2) - [\mathbb{E}(W)]^2 \\ &= \frac{5}{2} - 1^2 = \boxed{\frac{3}{2}}.\end{aligned}$$

Problem 4: Do participants find jobs more often?

For a randomly selected student, let $D = 1$ denote training participation and $H = 1$ employment. The table gives joint probabilities from an observational population.

	$D = 0$	$D = 1$	Total
$H = 0$	.30	.10	
$H = 1$	.30	.30	
Total			1

1. Complete the marginal totals. Find $\Pr(D = 1, H = 1)$.
2. Calculate $\Pr(H = 1 \mid D = 1)$, $\Pr(D = 1 \mid H = 1)$, and $\Pr(H = 1 \mid D = 0)$.
3. Find $\mathbb{E}(H \mid D = 1)$ and $\mathbb{E}(H \mid D = 0)$. Use iterated expectations to recover $\mathbb{E}(H)$.

Solution 4: Participation, employment, and conditioning

1.

	$D=0$	$D=1$	Total
$H=0$	.30	.10	.40
$H=1$	.30	.30	.60
Total	.60	.40	1

$$\Pr(D=1, H=1) = .30.$$

2.

$$\Pr(H=1 \mid D=1) = \frac{.30}{.40} = .75,$$

$$\Pr(D=1 \mid H=1) = \frac{.30}{.60} = .50,$$

$$\Pr(H=1 \mid D=0) = \frac{.30}{.60} = .50.$$

3. Because H is binary, $\mathbb{E}(H \mid D=d) = \Pr(H=1 \mid D=d)$. Therefore

$$\begin{aligned}\mathbb{E}(H) &= \mathbb{E}[\mathbb{E}(H \mid D)] \\ &= .75\Pr(D=1) + .50\Pr(D=0) = .75(.40) + .50(.60) = \boxed{.60}.\end{aligned}$$

Problem 5: What would independence look like?

Before training begins, an independent lottery assigns $A = 1$ with probability .40. Pre-programme employment H has probability $\Pr(H = 1) = .60$. Lottery assignment is independent of H.

1. Use independence to construct the full joint probability table of (A, H), including marginal totals.
2. Calculate $\Pr(H = 1 \mid A = 1)$, $\mathbb{E}(AH)$, and $\mathbb{E}(A)\mathbb{E}(H)$.

Solution 5: Independence before the programme

1.

	$A=0$	$A=1$	Total
$H=0$	.24	.16	.40
$H=1$	.36	.24	.60
Total	.60	.40	1

Every cell factorizes:

$$\Pr(A=0,H=0)=.60(.40)=.24,$$
$$\Pr(A=1,H=0)=.40(.40)=.16,$$
$$\Pr(A=0,H=1)=.60(.60)=.36,$$
$$\Pr(A=1,H=1)=.40(.60)=.24.$$

2.

$$\Pr(H=1\mid A=1)=\frac{.24}{.40}=.60=\Pr(H=1),\qquad \mathbb{E}(AH)=.24=.40(.60)=\mathbb{E}(A)\mathbb{E}(H).$$

Problem 6: How strongly do training and jobs move together?

Return to the observational population in Problem 4:

	$D = 0$	$D = 1$	Total
$H = 0$	.30	.10	
$H = 1$	.30	.30	
Total			1

1. Find $\mathbb{E}(D)$, $\mathbb{E}(H)$, $\text{Var}(D)$, $\text{Var}(H)$, and $\mathbb{E}(DH)$.
2. Calculate $\text{Cov}(D, H) = \mathbb{E}(DH) - \mathbb{E}(D)\mathbb{E}(H)$. Interpret its sign.
3. Find $\text{Corr}(D, H)$. Explain the role of standard deviations in this calculation.
4. Recode participation as $Q = 100D$. Find $\text{Cov}(Q, H)$ and $\text{Corr}(Q, H)$. What happens to correlation if employment is replaced with unemployment $U = 1 - H$?

Solution 6: Covariance from the lecture definition

1. Since D and H are binary,

$$\mathbb{E}(D) = .40, \qquad \mathbb{E}(H) = .60, \qquad \text{Var}(D) = .40(.60) = .24 = \text{Var}(H), \qquad \mathbb{E}(DH) = .30.$$

2. *Lecture 1, "Covariance"* defines covariance using deviations from the means. Let $\mu_D = \mathbb{E}(D)$ and $\mu_H = \mathbb{E}(H)$. Expanding that definition gives

$$\begin{aligned}
\text{Cov}(D, H) &= \mathbb{E}[(D - \mu_D)(H - \mu_H)] \\
&= \mathbb{E}[DH - \mu_D H - \mu_H D + \mu_D \mu_H] \\
&= \mathbb{E}(DH) - \mu_D \mathbb{E}(H) - \mu_H \mathbb{E}(D) + \mu_D \mu_H \\
&= \mathbb{E}(DH) - \mu_D \mu_H \\
&= \mathbb{E}(DH) - \mathbb{E}(D)\mathbb{E}(H).
\end{aligned}$$

The means are constants, so the cross terms reduce to the product of the means. Here,

$$\text{Cov}(D, H) = .30 - (.40)(.60) = \boxed{.06}.$$

Positive covariance indicates that participation and employment tend to move together.

Solution 6: Correlation and recoding (continued)

3.

$$\text{Corr}(D, H) = \frac{.06}{\sqrt{.24}\sqrt{.24}} = \frac{.06}{.24} = \boxed{.25}.$$

Dividing by standard deviations removes units and gives a scale-invariant measure of linear association.

Recoding the variables

4. For $Q = 100D$ and $U = 1 - H$,

$$\text{Cov}(Q, H) = 100(.06) = 6, \qquad \text{Corr}(Q, H) = .25,$$
$$\text{Cov}(D, U) = -\text{Cov}(D, H) = -.06, \qquad \text{Corr}(D, U) = -.25.$$

Positive scaling preserves correlation; reversing the employment indicator reverses its sign.

These associations alone do not identify a causal effect of training.

Problem 7: Can zero correlation hide a relationship?

A student's deviation from target work hours is $X \in \{-1, 0, 1\}$, with each value equally likely. The scheduling penalty index is $Z = X^2$.

1. List the possible pairs (X, Z) and their joint probabilities.
2. Calculate $\mathbb{E}(X)$, $\mathbb{E}(Z)$, $\text{Var}(X)$, and $\text{Var}(Z)$.
3. Find $\mathbb{E}(XZ)$, $\text{Cov}(X, Z)$, and $\text{Corr}(X, Z)$.
4. Compare $\Pr(Z = 0 \mid X = 0)$ with $\Pr(Z = 0)$. Are X and Z independent?
5. Explain why your answers about correlation and independence are compatible.

Solution 7: Zero correlation can hide dependence

1.

X	$Z = X^2$	Probability
-1	1	$1/3$
0	0	$1/3$
1	1	$1/3$

2.

$$\mathbb{E}(X) = 0, \qquad \mathbb{E}(Z) = \frac{2}{3},$$

$$\mathrm{Var}(X) = \frac{2}{3}, \quad \mathrm{Var}(Z) = \frac{2}{3} - \left(\frac{2}{3}\right)^2 = \frac{2}{9}.$$

3.

$$\mathbb{E}(XZ) = \frac{-1 + 0 + 1}{3} = 0,$$

$$\mathrm{Cov}(X, Z) = 0 - 0\left(\frac{2}{3}\right) = 0,$$

$$\mathrm{Corr}(X, Z) = \frac{0}{\sqrt{2/3}\sqrt{2/9}} = 0.$$

Yet conditioning changes the distribution: **4.**

$$\Pr(Z = 0 \mid X = 0) = 1 \neq \frac{1}{3} = \Pr(Z = 0).$$

5.

Interpretation

X and Z are dependent: X determines Z exactly. Correlation measures linear association and can be zero for a nonlinear relationship.

Problem 8: How risky is a two-part support payment?

Use the observational (D, H) distribution from Problem 4. You found

$$\mathbb{E}(D) = .40, \quad \mathbb{E}(H) = .60, \quad \mathrm{Var}(D) = \mathrm{Var}(H) = .24, \quad \mathrm{Cov}(D, H) = .06.$$

The university pays 100 euros for participating in training and 200 euros for being employed. Total support is $S = 100D + 200H$.

1. Compute $\mathbb{E}(S)$, $\mathrm{Var}(S)$, and its standard deviation. Include the covariance term.
2. How much would the variance be understated if covariance were incorrectly set to zero?
3. List the possible support payments and their probabilities. Verify the variance using $\mathbb{E}(S^2) - [\mathbb{E}(S)]^2$.

Solution 8: The risk of a two-part support payment

1. For $S = 100D + 200H$, expected support is

$$\mathbb{E}(S) = 100(.40) + 200(.60) = 160 \text{ euros.}$$

$$\begin{aligned}\text{Var}(S) &= 100^2\,\text{Var}(D) + 200^2\,\text{Var}(H) + 2(100)(200)\,\text{Cov}(D,H)\\ &= 2{,}400 + 9{,}600 + 2{,}400 = 14{,}400 \text{ euros}^2,\\ \text{SD}(S) &= \sqrt{14{,}400} = 120 \text{ euros.}\end{aligned}$$

2. Omitting covariance gives 12,000 euros^2, understating variance by 2,400 euros^2. **3.**

s (euros)	0	100	200	300
$\Pr(S = s)$	.30	.10	.30	.30

$$\mathbb{E}(S^2) = .10(100^2) + .30(200^2) + .30(300^2) = 40{,}000, \qquad \text{Var}(S) = 40{,}000 - 160^2 = 14{,}400.$$

Problem 9: Which earnings survey can we trust?

The target is the mean hourly wage among employed students at UPF. Consider four plans:

A. Sample student records independently with replacement from the complete target-population list.

B. Post a voluntary wage survey in a job-training group chat.

C. Randomly select five employers and interview all UPF students working for them.

D. Randomly select 100 distinct students from the complete target-population list, without replacement.

1. Which plan directly gives independent, identically distributed draws? Discuss selection and dependence in the other plans.
2. What could nonresponse do to an otherwise sound design? Does increasing the number of voluntary replies remove selection bias?

Solution 9: Choosing the survey design

1.

Plan	Implication for the earnings survey
A	With complete responses and consistent wage measurement, independent selection with replacement gives i.i.d. draws from the target list's wage distribution.
B	Volunteers in the training chat can differ systematically from all employed UPF students. The respondent distribution need not equal the target distribution.
C	This is a cluster sample: students at the same employer may share wage shocks. Independence can fail; individual inclusion probabilities also require care.
D	Sampling distinct students without replacement gives equal marginal distributions, but the draws are dependent in a finite population. An i.i.d. approximation can be reasonable for a small sampling fraction.

2.

- Wage-related nonresponse can change the respondent distribution under any plan.
- More voluntary replies increase the amount of data; they do not automatically remove selection bias.

Problem 10: What can two employment responses tell us?

Employment indicators H_1 and H_2 are independent draws from a population with $\Pr(H_i = 1) = .60$. Define

$$\overline{H} = \frac{H_1 + H_2}{2}.$$

1. List the four ordered samples (H_1, H_2) and their probabilities.
2. Construct the exact sampling distribution of $\overline{H}$.
3. Calculate $\mathbb{E}(\overline{H})$ and $\mathrm{Var}(\overline{H})$ directly from that distribution.
4. Check your answers using $\mathbb{E}(\overline{H}) = \mu_H$ and $\mathrm{Var}(\overline{H}) = \sigma_H^2/n$.
5. How does the sampling distribution of an average differ from the distribution of one student's employment status?

Solution 10: The exact sampling distribution

1. Independence gives the probability of each ordered pair as a product:

(H_1, H_2)	Probability	$\overline{H}$
$(0,0)$	$.4 \times .4 = .16$	0
$(0,1)$	$.4 \times .6 = .24$	.5
$(1,0)$	$.6 \times .4 = .24$	.5
$(1,1)$	$.6 \times .6 = .36$	1

2. Collect equal values of the average:

h	$\Pr(\overline{H} = h)$
0	.16
.5	$.24 + .24 = .48$
1	.36

3.

$$\mathbb{E}(\overline{H}) = 0(.16) + .5(.48) + 1(.36) = .60,$$
$$\mathbb{E}(\overline{H}^2) = 0^2(.16) + .5^2(.48) + 1^2(.36) = .48,$$
$$\text{Var}(\overline{H}) = .48 - (.60)^2 = .12.$$

4. The i.i.d. formula confirms $\mathbb{E}(\overline{H}) = .60 = \mu_H$ and $\text{Var}(\overline{H}) = .12 = \sigma_H^2/2 = .60(.40)/2$.

5. One student's H takes values $0, 1$; the statistic $\overline{H}$ takes values $0, .5, 1$ across repeated samples of size two.

Problem 11: How precise is a survey's average wage?

For this problem and Problems 12–13, model hourly wages $Y_1, \ldots, Y_n$ as i.i.d. draws with population mean $\mu_Y = 15$ euros and standard deviation $\sigma_Y = 6$ euros.

1. Derive $\mathbb{E}(\overline{Y})$ and $\mathrm{Var}(\overline{Y})$. Identify where independence is used.
2. Compute the variance and standard deviation of $\overline{Y}$ for $n = 25$ and $n = 100$.
3. Does quadrupling the sample size divide the standard deviation by four?
4. How large must n be for the standard deviation of $\overline{Y}$ to be at most .50 euros?
5. If students' wages are positively correlated, what missing terms would make the i.i.d. variance formula too small?

Solution 11: Sample size and earnings precision

1.

$$\mathbb{E}(\overline{Y}) = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}(Y_i) = 15, \qquad \mathrm{Var}(\overline{Y}) = \frac{1}{n^2}\left[\sum_{i=1}^{n}\mathrm{Var}(Y_i) + \sum_{i\neq j}\mathrm{Cov}(Y_i, Y_j)\right] = \frac{36n + 0}{n^2} = \frac{36}{n}.$$

Independence makes the cross-observation covariances zero. Therefore $\mathrm{SD}(\overline{Y}) = 6/\sqrt{n}$.

2.

Sample size	$\mathrm{Var}(\overline{Y})$ (euros squared)	$\mathrm{SD}(\overline{Y})$ (euros)
25	$36/25 = 1.44$	1.20
100	$36/100 = .36$	.60

- **3.** Quadrupling n **halves** the SD: $6/\sqrt{4n} = \frac{1}{2}(6/\sqrt{n})$.
- **4.** For SD at most .50: $6/\sqrt{n} \leq .50 \iff \sqrt{n} \geq 12 \iff n \geq 144$.
- **5.** Positive covariances add $n^{-2}\sum_{i\neq j}\mathrm{Cov}(Y_i, Y_j)$ to the variance. Omitting them understates uncertainty.

Problem 12: Will a larger survey find the true mean?

Continue with the i.i.d. wage population: $\mu_Y = 15$ euros and $\sigma_Y^2 = 36$ euros squared.

1. State the law of large numbers for $\overline{Y}_n$ using convergence in probability.
2. For a fixed tolerance $\varepsilon = .50$ euros, complete the claim

$$\Pr(|\overline{Y}_n - 15| < .50) \longrightarrow _____.$$

3. If the wage population has a continuous distribution, is $\Pr(\overline{Y}_n = 15)$ positive at a finite n? Explain.
4. Does the law guarantee that every additional response makes the realized sample mean closer to 15?
5. Why does a large voluntary sample with selection bias require a separate argument?

Solution 12: What the law of large numbers says

1. For i.i.d. wages with finite mean, the law of large numbers gives

$$\overline{Y}_n \xrightarrow{p} 15 \quad \Longleftrightarrow \quad \Pr(|\overline{Y}_n - 15| < \varepsilon) \longrightarrow 1 \quad \text{for every fixed } \varepsilon > 0.$$

2. The blank for the tolerance .50 euros is therefore **1**.

- **3.** With independent draws from a continuous wage distribution, $\overline{Y}_n$ is continuous, so $\Pr(\overline{Y}_n = 15) = 0$ at every finite n.

- **4.** The convergence statement concerns probabilities of *neighborhoods* of 15. It does not require exact equality or a realized average that improves after every response.

- **5.** A selected sample can obey an LLN for its own population. For example, if volunteer wages are i.i.d. with mean $\mu_{\text{vol}} \neq 15$, then

  $$\overline{Y}_{n,\text{vol}} \xrightarrow{p} \mu_{\text{vol}},$$

  so a large sample alone does not establish convergence to the target mean.

Problem 13: Is an unusually high survey mean surprising?

Hourly wages are i.i.d., with $\mu_Y = 15$ euros and $\sigma_Y = 6$ euros. The population distribution is right skewed with finite variance. A survey samples $n = 100$ students. Assume the CLT approximation is adequate at this sample size.

1. Use the central limit theorem to approximate the distribution of $\overline{Y}$. State whether the second normal parameter is a variance or a standard deviation.
2. Standardize and approximate $\Pr(\overline{Y} > 16.20)$.
3. Approximate $\Pr(13.80 \leq \overline{Y} \leq 16.20)$.
4. Would the normal approximation be exact for this population? What changes if the population itself is normal? Is the approximation automatically reliable at $n = 4$?

Use $\Phi(2) = .9772$ and $\Phi(-2) = .0228$, where Φ is the standard normal c.d.f.

Solution 13: A high average under the CLT

1. At $n = 100$, the assumed CLT approximation gives

$$\overline{Y} \overset{a}{\sim} \mathcal{N}\left(15, \frac{36}{100}\right) = \mathcal{N}(15, .36), \qquad Z = \frac{\overline{Y} - 15}{.60} \overset{a}{\sim} \mathcal{N}(0, 1).$$

The second normal parameter is the **variance**; the SD is .60 euros. **2.**

$$\Pr(\overline{Y} > 16.20) \approx \Pr\left(Z > \frac{16.20 - 15}{.60}\right) = 1 - \Phi(2) = .0228.$$

3.

$$\Pr(13.80 \leq \overline{Y} \leq 16.20) \approx \Pr(-2 \leq Z \leq 2) = \Phi(2) - \Phi(-2) = .9544.$$

- **4.** For the right-skewed population these are approximate probabilities; if wages themselves are normal, the sample mean is *exactly* normal at every n. Finite variance does not guarantee an accurate approximation at $n = 4$; skewness and rare outliers affect finite-sample accuracy.

Problem 14: Which wage estimator should we use?

Let $Y_1, \ldots, Y_n$ be i.i.d. with unknown mean μ_Y and variance 36. Consider

$$A_n = \overline{Y}_n, \qquad B_n = Y_1, \qquad C_n = \overline{Y}_n + 1, \qquad T_n = \frac{n-1}{n}\overline{Y}_n.$$

1. For each rule, find its expectation, bias, and variance. Which are unbiased for all μ_Y?
2. Which rules are consistent for μ_Y? Explain what happens as $n \to \infty$.
3. At $n = 100$, which unbiased rule is more efficient? Does a smaller variance alone make a biased rule preferable?
4. For the observed wages $(9, 12, 15, 18, 21)$, compute the four estimates with $n = 5$. Distinguish each estimator from its realized estimate.

Solution 14: Bias and consistency of four rules

1. Let $a_n = (n-1)/n$. Linearity of expectation and $\text{Var}(cX) = c^2 \text{Var}(X)$ give:

Estimator	Expectation	Bias for μ_Y	Variance
$A_n = \overline{Y}_n$	μ_Y	0	$36/n$
$B_n = Y_1$	μ_Y	0	36
$C_n = \overline{Y}_n + 1$	$\mu_Y + 1$	1	$36/n$
$T_n = a_n \overline{Y}_n$	$a_n \mu_Y$	$-\mu_Y/n$	$a_n^2 \, 36/n$

Thus A_n **and** B_n **are unbiased for every** μ_Y. **2.**

- $A_n \xrightarrow{p} \mu_Y$ by the LLN: it is consistent.
- $B_n = Y_1$ retains the same nondegenerate distribution as n grows: it is inconsistent.
- $C_n \xrightarrow{p} \mu_Y + 1$: it is inconsistent for μ_Y.
- $a_n \to 1$ and $\overline{Y}_n \xrightarrow{p} \mu_Y$, so $T_n \xrightarrow{p} \mu_Y$: it is consistent, although generally biased at finite n.

Solution 14: Precision and realized estimates (continued)

3. At $n = 100$: compare the unbiased rules.

$$\text{Var}(A_{100}) = \frac{36}{100} = .36 \quad < \quad \text{Var}(B_{100}) = 36.$$

The sample mean is more efficient than the first observation among these unbiased rules. Shrinkage reduces variance but introduces bias:

$$\text{Var}(T_{100}) = .99^2(.36) = .352836, \qquad \mathbb{E}(T_{100}) - \mu_Y = -.01\mu_Y.$$

A smaller variance alone does not establish that a biased estimator is more accurate.

4. After observing $(9, 12, 15, 18, 21)$, use $n = 5$:

$$\bar{y} = \frac{9 + 12 + 15 + 18 + 21}{5} = 15,$$

rule	A_5	B_5	C_5	T_5
estimate	15	9	16	$\frac{4}{5}(15) = 12$

An *estimator* is the sample-based rule, random before observation; an *estimate* is the numerical value produced by the observed sample.

Problem 15: How precise can an unbiased wage estimate be?

Now suppose $Y_1, \ldots, Y_{100}$ are i.i.d. normal with unknown mean μ_Y and *known* variance $\sigma_Y^2 = 36$. You may use

$$\mathcal{I}_n(\mu_Y) = \frac{n}{\sigma_Y^2}, \qquad \operatorname{Var}(\widehat{\mu}_Y) \geq \frac{1}{\mathcal{I}_n(\mu_Y)} \quad \text{for unbiased estimators.}$$

1. Calculate the Fisher information and the Cramér–Rao lower bound.
2. Show that the sample mean reaches this bound. What does this establish about its efficiency in this model?
3. Can another unbiased estimator have variance .30 under the same assumptions?
4. The shrinkage rule $T_{100} = .99\overline{Y}$ has variance below that of $\overline{Y}$. Why is this compatible with the bound?

Solution 15: The normal-model precision bound

1. For $n = 100$ independent normal wages with known variance 36,

$$\mathcal{I}_{100}(\mu_Y) = \frac{100}{36} = \frac{25}{9}, \qquad \frac{1}{\mathcal{I}_{100}(\mu_Y)} = \frac{9}{25} = .36.$$

The Cramér–Rao bound therefore requires $\mathrm{Var}(\widehat{\mu}_Y) \geq .36$ for an estimator unbiased throughout this parameter space. **2.**

$$\mathbb{E}(\overline{Y}) = \mu_Y, \qquad \mathrm{Var}(\overline{Y}) = \frac{36}{100} = .36.$$

The sample mean attains the bound: it is **efficient among unbiased estimators in this normal model**. **3.** An unbiased estimator with variance .30 would violate the bound.

4. For $T_{100} = .99\overline{Y}$,

$$\mathbb{E}(T_{100}) = .99\mu_Y, \qquad \mathrm{Bias}(T_{100}) = -.01\mu_Y, \qquad \mathrm{Var}(T_{100}) = .352836.$$

Its smaller variance is compatible with the bound because it is not unbiased for every μ_Y.

Problem 16: Why give every wage the same weight?

Four i.i.d. wages have mean μ_Y and variance 36. Normality is *not* assumed. Compare

$$M = \frac{Y_1 + Y_2 + Y_3 + Y_4}{4}, \qquad L = \frac{1}{2}Y_1 + \frac{1}{6}(Y_2 + Y_3 + Y_4).$$

1. Show that both estimators are linear and unbiased. Compare their variances.
2. For fixed weights, when is $\widehat{\mu} = \sum_{i=1}^{n} w_i Y_i$ unbiased for every μ_Y? Derive its variance under i.i.d. sampling.
3. Under the unbiasedness restriction, use

$$\sum_{i=1}^{n}(w_i - 1/n)^2 = \sum_{i=1}^{n} w_i^2 - 1/n \geq 0$$

   to find the minimum possible variance and the weights attaining it.
4. Explain each word in "Best Linear Unbiased Estimator" for the sample mean.

Solution 16: Comparing two wage averages

1. Both rules are fixed weighted sums, and both sets of weights sum to one:

$$\mathbb{E}(M) = \frac{1}{4}(4\mu_Y) = \mu_Y, \qquad \mathbb{E}(L) = \left(\frac{1}{2} + 3 \cdot \frac{1}{6}\right)\mu_Y = \mu_Y.$$

Independence removes covariance terms, so

$$\mathrm{Var}(M) = 36\left[4\left(\frac{1}{4}\right)^2\right] = 9,$$

$$\mathrm{Var}(L) = 36\left[\left(\frac{1}{2}\right)^2 + 3\left(\frac{1}{6}\right)^2\right] = 12.$$

Thus M is more efficient than L among these two unbiased estimators.

2. For any fixed weights $w_1, \ldots, w_n$ under the stated i.i.d. model,

$$\mathbb{E}\left(\sum_{i=1}^n w_i Y_i\right) = \mu_Y \sum_{i=1}^n w_i, \qquad \mathrm{Var}\left(\sum_{i=1}^n w_i Y_i\right) = 36\sum_{i=1}^n w_i^2.$$

Unbiasedness for *every* μ_Y requires $\sum_{i=1}^n w_i = 1$.

Solution 16: Why equal weights are BLUE (continued)

3. Impose the unbiasedness restriction $\sum_{i=1}^{n} w_i = 1$. Then

$$0 \le \sum_{i=1}^{n} \left(w_i - \frac{1}{n} \right)^2 = \sum_{i=1}^{n} w_i^2 - \frac{2}{n} \sum_{i=1}^{n} w_i + \frac{n}{n^2}$$
$$= \sum_{i=1}^{n} w_i^2 - \frac{1}{n}.$$

Consequently,

$$\operatorname{Var}\left(\sum_{i=1}^{n} w_i Y_i \right) = 36 \sum_{i=1}^{n} w_i^2 \ \ge \ \frac{36}{n},$$

with equality precisely when $w_i = 1/n$ for every i. The attaining estimator is $\overline{Y}$.

- **4.** **BLUE** means Best Linear Unbiased Estimator: best is smallest variance within the linear unbiased class, linear is a fixed weighted sum, and unbiased means the expectation equals the population mean.

Normality is unnecessary; i.i.d. sampling supplies the common mean, variance, and zero covariances used here.

Problem 17: Which wage best fits all observations?

Five observed hourly wages, in euros, are

$$(y_1, y_2, y_3, y_4, y_5) = (9, 12, 15, 18, 21).$$

Choose one number m to minimize the sum of squared deviations,

$$Q(m) = \sum_{i=1}^{5} (y_i - m)^2.$$

1. Differentiate $Q(m)$, solve the first-order condition, and verify that it is a minimum.
2. Compute $Q(12)$ and $Q(15)$. What is the minimizing estimate?
3. Derive $Q(m) = Q(\bar{y}) + 5(m - \bar{y})^2$. What does this identity show about uniqueness?
4. State the least-squares estimator before observing the data. Which additional sampling conditions justify calling it unbiased and BLUE for the population mean?

Solution 17: Least squares chooses the sample mean

1. The observed wages satisfy $\sum y_i = 75$ and $\bar{y} = 15$. Hence

$$Q'(m) = -2\sum_{i=1}^{5}(y_i - m) = 10m - 150 = 0 \quad \Longrightarrow \quad m = 15, \qquad Q''(m) = 10 > 0.$$

2. Direct comparison gives

$$Q(12) = 9 + 0 + 9 + 36 + 81 = 135, \qquad Q(15) = 36 + 9 + 0 + 9 + 36 = 90.$$

The minimizing wage

The least-squares estimate is 15 euros per hour. The positive second derivative verifies that this stationary point is a minimum.

Solution 17: Uniqueness and sampling properties (continued)

3. Since $\sum_{i=1}^{5}(y_i - \bar{y}) = 0$, expanding the square yields

$$\begin{aligned} Q(m) &= \sum_{i=1}^{5}[(y_i - \bar{y}) + (\bar{y} - m)]^2 \\ &= Q(\bar{y}) + 2(\bar{y} - m)\underbrace{\sum_{i=1}^{5}(y_i - \bar{y})}_{0} + 5(\bar{y} - m)^2 \\ &= 90 + 5(m - 15)^2. \end{aligned}$$

This is uniquely minimized at $m = 15$.

4. Before observing data, the least-squares estimator is $\widehat{\mu} = \overline{Y}$. The minimization is algebraic. Common finite means make $\overline{Y}$ unbiased; independent observations with a common finite positive variance also make it BLUE. The course's i.i.d. assumptions suffice for both properties.